Hussain Mustansir

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Final-year Computer Science student with hands-on experience in AI and machine learning. Skilled in developing NLP, computer vision, and predictive models using Python, TensorFlow, PyTorch, and Transformer architectures. Experienced in data preprocessing, model training, evaluation, and deployment on cloud platforms. Strong foundation in deep learning, neural networks, and algorithm optimization, with published research in legal judgment prediction and network security.

Education

Bachelor of Science in Computer Science

Habib University, Karachi, Pakistan | 2022 – 2026 (Expected)

Relevant Coursework: Introduction To Deep Learning, Introduction to Large Models, Software Quality & Assurance, Enterprise Software Development, Cryptography & Network Security, IoT & Cloud Computing

Technical Skills

- **Machine Learning & AI:** Python, TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers, NLP, Computer Vision, CNNs, Data Preprocessing, Model Evaluation & Optimization
- **Data & Databases:** Pandas, NumPy, SQL, MongoDB, PostgreSQL, API Data Integration
- **DevOps & Cloud:** Docker, CI/CD pipelines (GitHub Actions), AWS EC2, Vercel Deployment
- **Tools & Libraries:** Jupyter, Weights & Biases (W&B), OpenCV, Tesseract, BeautifulSoup, Postman, Selenium
- **Programming & Backend**: JavaScript (Node.js, Express.js), Python, REST API Integration, JWT authentication
- **Frontend & Web:** React.js, Next.js, JavaScript/TypeScript, HTML/CSS, Responsive UI Development

Academic Projects & Research Publication

- Prediction Model For Legal Judgments | Research Publication
  - Built an AI model using Transformer-based NLP models (BART, T5 Base, Distil GPT-2, Facebook-opt-125m) to predict criminal case judgments under Pakistani law.
  - Processed and trained on judgments using Python using libraries BeautifulSoup, Tesseract, and Pandas.
  - Tracked model performance with Weights & Biases (W&B) and evaluated accuracy using ROUGE-1 scores, with BART achieving the best results.

- **Deep Learning Model for Traffic Sign Recognition | Independent Project**
  - Built a CNN-based image classification model using TensorFlow and OpenCV for real-time traffic sign detection.
  - Achieved 96% accuracy on GTSRB dataset after applying data augmentation and dropout regularization.

- **AI Study Buddy | Software Engineering**
  - Developed a full-stack web application to help students manage study plans, notes, and tasks with AI assistance.
  - Implemented dashboards, task management components, and updates using React hooks and state management.
  - Built with React.js (frontend) and Node.js, Express.js, Python (backend) with MongoDB, deployed on Vercel.

- **Karavaan | Final Year Project | Spursol**
  - Built frontend components for communities, events, fundraisers, messaging, and SOS systems using React.js.
  - Integrated OTP, payment gateway, notifications, and mapping services into a unified backend architecture.
  - Designed a large-scale relational schema supporting entities with transactional safety and high performance.
  - Contributed to frontend integration and API testing using Postman, improving development speed and reliability.

- **Security at Network Layer – IPSec and VPNs | Research Publication**
  - Explored network security solutions at the network layer, focusing on IPSec protocols and WireGuard.
  - Analyzed VPN architectures, encryption methods and protocols for secure communication.
  - Implemented and tested a WireGuard VPN server on AWS EC2 with secure client connectivity, evaluating performance and security benefits over IPSec.